

Abortion & Post-Abortion Care

Tutorial

Week 1 · Foundations, ANC & early pregnancy

Gynaecology

Dr Nassimu

Abortion is the loss or termination of a pregnancy before fetal viability (28 weeks in Ugandan practice). Classify it by bleeding, pain, the cervical os, uterine size and tissue passed. Resuscitate first, then evacuate the uterus with MVA or misoprostol, treat sepsis aggressively, and never let a woman leave without post-abortion contraception.

Must know before the ward round

1. Classify by the os and the uterus: os closed + size = dates means threatened; os open means inevitable or incomplete; os closed + small uterus + bleeding settled means complete; os closed + fetus dead and retained means missed.
2. Stability first: shock index (pulse ÷ systolic BP) of 0.9 or more is abnormal and 1.4 or more is critical. Two large-bore cannulae, 1 L crystalloid, then reassess.
3. The speculum is the key examination: look at the os, remove products from the os with sponge forceps, and look for tears, burns, tenaculum marks or foreign bodies.
4. Incomplete abortion with uterine size under 14 weeks: MVA, or misoprostol 600 µg orally once, or 400 µg sublingually once (WHO 2022). Sharp curettage (D&C) is obsolete.
5. Uterus 14 weeks or more: misoprostol 400 µg sublingual, vaginal or buccal every 3 hours until expulsion, then check for retained tissue.
6. Missed miscarriage on ultrasound: CRL 7 mm or more with no heartbeat, or mean sac diameter 25 mm or more with no embryo. Below these cut-offs, rescan in 7–14 days.
7. Missed abortion under 14 weeks (WHO 2022): mifepristone 200 mg orally, then misoprostol 800 µg 1–2 days later; misoprostol 800 µg alone if no mifepristone.
8. Septic abortion: resuscitate, give IV antibiotics at once (ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily + metronidazole 500 mg IV 8-hourly), then evacuate within hours. Retained infected tissue is the source, so evacuation is the source control.
9. Bradycardia and hypotension out of proportion to the blood loss mean cervical shock: remove the products from the os with sponge forceps.
10. Closed os + empty uterus + positive pregnancy test with no previously confirmed intrauterine pregnancy is a pregnancy of unknown location (possible ectopic), not a complete abortion.
11. After unsafe abortion give tetanus toxoid 0.5 mL IM (± tetanus immunoglobulin 250 IU); give anti-D within 72 h to Rh-negative women after surgical evacuation or loss at 12 weeks or more.
12. Fertility returns within 2 weeks of an abortion. Offer contraception before discharge; almost every method, including an IUD, can start the same day unless there is sepsis. PAC is legal in Uganda and must never be delayed.

- 1 In a nutshell
- 2 Definitions & classification
- 3 Epidemiology & Uganda context
- 4 Pathophysiology
- 5 Clinical acumen
- 6 Clinical workup

- 7 Differential diagnosis
- 8 Investigations
- 9 Management
- 10 Complications
- 11 Ward-round & exam pearls
- 12 Mnemonics & memory aids

In a nutshell

- **Abortion** is the ending of a pregnancy before the fetus can survive outside the uterus. Ugandan clinical practice uses **before 28 weeks**; WHO uses **before 22 weeks or under 500 g** for international comparison.
- Most spontaneous abortions (**miscarriages**) happen in the first trimester and are due to **chromosomal abnormalities**. Most deaths, however, come from **unsafe induced abortion**, through haemorrhage, sepsis and genital or bowel trauma.
- In Uganda, **abortion complications are a leading cause of gynaecological admission** and a consistent top cause of maternal death in MPDSR (maternal and perinatal death surveillance and response) reviews.
- The bedside question is always "**Is she stable?**", then "**What type of abortion is it?**" (the speculum look at the os answers most of this), then "**Is there sepsis, or an injury?**"
- Treatment is **uterine evacuation**. Use **manual vacuum aspiration (MVA)** or **misoprostol** below 14 weeks' uterine size, and **misoprostol** from 14 weeks, plus antibiotics when infected.
- **Post-abortion care (PAC)** is a package, not just a procedure: emergency treatment, counselling, **family planning before discharge**, other reproductive health services, and community links.
- **The one thing never to forget:** a woman who is bleeding in early pregnancy with a **closed os and an empty uterus** may have an **ectopic pregnancy**. Do not label it "complete abortion" without evidence.

Definitions & classification

Type	Bleeding	Pain	Cervical os	Uterus vs dates	Tissue passed	Ultrasound
Threatened	Light to moderate	Mild or none	Closed	Equal to dates	None	Viable intrauterine pregnancy (IUP)
Inevitable	Heavy	Cramping, often strong	Open	Equal to or less than dates	None yet (membranes may have ruptured)	Sac low in the uterus or in the cervix
Incomplete	Heavy, ongoing	Cramping	Open; products may sit in the os	Less than dates	Partial	Retained products (heterogeneous material)
Complete	Settling or minimal	Settling	Closed	Small, firm	All (sac seen)	Empty uterus, thin endometrium
Missed (silent miscarriage)	Absent or brownish spotting	None	Closed	Less than dates	None	Embryo without heartbeat, or empty sac
Septic	Any, often offensive	Lower abdominal or pelvic pain	Usually open	Tender, bulky	Often incomplete	Retained products, with or without free fluid

Other terms you will meet:

- **Spontaneous abortion (miscarriage):** no deliberate intervention.
- **Induced abortion:** deliberately ended, by medication or by a procedure.
- **Unsafe abortion (WHO):** a procedure to end a pregnancy done by someone without the necessary skills, or in an environment that does not meet minimal medical standards, or both. Examples include sticks, herbs, cassava stems, bicycle spokes, overdoses of chloroquine or other drugs, and "washing" with detergents.

- **Recurrent miscarriage:** classically 3 or more consecutive losses (older RCOG definition, still used in much teaching). ESHRE and the 2023 RCOG update now use 2 or more.
- **Early pregnancy loss** (ultrasound terms): a **missed miscarriage** is diagnosed when the crown–rump length (CRL) is **7 mm or more with no heartbeat**, or the mean sac diameter is **25 mm or more with no embryo**. Below these cut-offs, **rescan in 7–14 days** rather than acting.

UGANDA CONTEXT

Ugandan clinical teaching and the Uganda Clinical Guidelines define abortion as pregnancy loss **before 28 weeks**. Research and WHO statistics often use 22 weeks. A loss at 24 weeks is therefore still managed on the gynaecology (abortion) pathway in most Ugandan hospitals, while it would be a "late fetal loss" or "stillbirth" elsewhere.

Epidemiology & Uganda context

- **10–20% of clinically recognised pregnancies miscarry**, and about 80% of these losses happen before 12 weeks. The main cause is **aneuploidy** (an abnormal number of chromosomes), especially autosomal trisomy.
- **Risk factors for spontaneous loss:** maternal age over 35, previous miscarriage, uncontrolled diabetes or thyroid disease, antiphospholipid syndrome, **malaria** (a febrile illness that inflames the placenta), **syphilis** (mainly later losses), uterine anomalies and fibroids, cervical insufficiency (painless second-trimester loss), smoking and alcohol.
- **Unsafe abortion in Uganda:** national estimates (Prada et al., 2016) put induced abortions at roughly **300,000 a year**, with close to **100,000 women treated for complications** each year. Most of those women are young, poor, rural, unmarried students, or have an unmet need for contraception.
- **Determinants of septic abortion in Ugandan teaching hospitals** include rural residence, no income, induced abortion, evacuation attempted at home or at a lower-level unit, and no use of contraception.
- **Facility realities:**
 - PAC with **misoprostol and MVA** is meant to be available at HC III and HC IV, and Ugandan trials show that **midwives achieve the same results as doctors** with misoprostol.
 - In practice, MVA kits may be missing or unprocessed, misoprostol is out of stock, blood is short, and women arrive late because of fear of stigma, police or cost.
 - An audit in a rural Ugandan hospital found **missing vital signs, poor fluid resuscitation of shocked women, and delayed evacuation of septic abortion**. These are exactly the gaps you can fix as a clerk.

UGANDA CONTEXT

Legal context (be precise and non-judgemental). Article 22(2) of the Constitution says no one may terminate the life of an unborn child "except as may be authorised by law". The Penal Code criminalises procuring an abortion but protects a surgical operation done in good faith to preserve the mother's life. The MoH National Policy Guidelines and Service Standards for Sexual and Reproductive Health allow termination in specific situations, for example serious maternal illness, rape, incest or defilement. The 2015 MoH standards on reducing unsafe abortion were withdrawn soon after release, so the policy space is contested. **What is not contested: post-abortion care is legal, lifesaving and a patient right.** Treat every woman the same, whatever the cause, keep her information confidential, and never delay care to interrogate her.

Pathophysiology

Why do most early pregnancies fail? About half of first-trimester miscarriages have a chromosomal abnormality: trisomy (commonest is trisomy 16), then 45,X, then triploidy. The embryo stops developing and hCG and progesterone fall. The **decidua (the pregnancy-altered endometrium) breaks down**, and prostaglandins are released, causing **cramping and cervical opening**. What you see clinically is the uterus trying to expel its contents.

Why does incomplete abortion bleed so much?

- After a pregnancy loss, bleeding normally stops through **myometrial contraction**. The "living ligatures" of the uterine muscle squeeze the spiral arteries shut.
- **Retained tissue keeps the uterus from contracting fully** and holds the placental bed open, so bleeding continues.
- The fix is therefore **mechanical**: empty the uterus with MVA, or chemically make it contract with misoprostol. Blood transfusion alone never solves the problem.

Cervical shock. Products stuck in the cervical canal stretch the cervix, which is richly supplied by parasympathetic (vagal) fibres. The result is a **vasovagal reflex**: bradycardia, hypotension and pallor, **out of proportion to the visible blood loss**. Removing the tissue from the os with sponge forceps reverses it within minutes.

Septic abortion.

- Instruments or foreign bodies carry vaginal, bowel and soil organisms into a **necrotic, blood-filled uterus**, which is a perfect culture medium.
- Organisms are **polymicrobial**: *E. coli* and other Gram-negative rods, anaerobes (*Bacteroides*, *Peptostreptococcus*), streptococci (including group A), and ***Clostridium perfringens*** (gas gangrene, haemolysis, bronze or jaundiced skin, renal failure). *Clostridium tetani* also matters, so give **tetanus prophylaxis**.
- Infection spreads from **endometritis → myometritis → parametritis → pelvic peritonitis / abscess → septicaemia → septic shock and DIC** (disseminated intravascular coagulation).
- Antibiotics cannot penetrate dead tissue well, so **evacuation is source control**. This is why you evacuate promptly after starting antibiotics, rather than "cooling her off" for days.
- Unsafe procedures can **perforate the uterus and injure bowel or bladder**. Suspect this with peritonitis, free gas under the diaphragm, or bowel or fat seen in the vagina or aspirate.

How the drugs work (and why that explains their side-effects).

- **Misoprostol** is a **prostaglandin E1 analogue**. It makes the myometrium contract and softens (ripens) the cervix. It is cheap, heat-stable and active by mouth.
 - Side-effects are the prostaglandin effects: **shivering, fever, diarrhoea, nausea and cramps**.
 - The **sublingual route** gives the fastest, highest peak. The **vaginal route** gives a longer effect with fewer side-effects.
- **Mifepristone** is a **progesterone receptor blocker**. Progesterone keeps the decidua alive and the uterus quiet, so blocking it causes **decidual necrosis, cervical softening and a uterus much more sensitive to prostaglandins**. That is why the combination (mifepristone first, then misoprostol 24–48 hours later) is more effective than misoprostol alone.
- **Oxytocin** works poorly in early pregnancy because the uterus has **few oxytocin receptors before about 20 weeks**. Prostaglandins, not oxytocin, are the drugs of choice for early losses.

Focused history

Question	Why it matters
LMP, cycle regularity, any early scan	Dates the pregnancy. Uterine size smaller than dates suggests incomplete or missed abortion. Size far larger than dates suggests a molar pregnancy , twins or fibroids.
How much bleeding? (pads soaked per hour, clots, "flooding")	Soaking 1 pad or more in under an hour , or passing large clots, means heavy bleeding. Look for shock.
Has any tissue passed? (ask what it looked like; did she bring it?)	Passed tissue suggests incomplete or complete abortion. Inspect anything she brought.
Pain: crampy central, or one-sided and constant? Shoulder-tip pain? Fainting?	Crampy midline pain means the uterus is expelling. Unilateral pain, shoulder-tip pain or collapse means ectopic pregnancy until proven otherwise.
Fever, rigors, offensive discharge	Sepsis
"Did anything happen before the bleeding?" Asked gently: any medicine, herb, insertion, or visit to anyone to "remove" the pregnancy?	Interference raises the risk of sepsis, perforation, retained foreign body, tetanus and drug toxicity. Ask privately, promise confidentiality, and explain it is to treat her safely, not to report her.
Was this pregnancy planned? Current contraception?	Opens PAC family-planning counselling. Unintended pregnancy is common after abortion.
Past obstetric history, previous miscarriages, previous CS	Recurrent loss work-up. A previous CS needs caution with high-dose misoprostol later in pregnancy (scar rupture).
Rhesus group if known; HIV status; sickle cell	Anti-D; PMTCT/ART linkage; anaemia risk
Medical illness (diabetes, hypertension, epilepsy, asthma, bleeding disorder)	Choice of drug and anaesthesia
Tetanus immunisation status	Tetanus toxoid ± tetanus immunoglobulin after unsafe abortion

Examination

- General:** pallor, level of consciousness, temperature, **pulse, BP, respiratory rate, capillary refill, urine output**. Calculate the **shock index** (pulse ÷ systolic BP): **0.9 or more is worrying, 1.4 or more is critical**.
- Abdomen:**
 - Is the uterus palpable (roughly more than 12 weeks)? Measure the fundal height.
 - Check for **guarding, rigidity, rebound** (peritonitis suggests perforation, pelvic abscess or ectopic) and **distension**, and listen for absent bowel sounds (ileus, bowel injury).
- Speculum (the single most useful step):**
 - Is the **os open or closed**?
 - Are **products of conception (POC) in the os**? Remove them now with sponge forceps.
 - Note the **source and amount of bleeding** and any **offensive discharge**.
 - Look for **lacerations, burns or foreign bodies** (e.g. a stick or catheter) and **tenaculum marks on the cervix**, which are evidence of instrumentation.
- Bimanual:**
 - Uterine size vs dates; consistency.
 - Cervical motion tenderness** (infection or ectopic).
 - Adnexal mass or tenderness** (ectopic, tubo-ovarian abscess).

- A **boggy pouch of Douglas** (blood, pus).

Red flags

RED FLAG

- **Shock:** pulse over 110, systolic BP under 90, shock index 1 or more, confusion, cold peripheries. Resuscitate before anything else.
- **Fever of 38 °C or more, rigors, foul discharge, uterine tenderness.** This is septic abortion: start antibiotics within the hour.
- **Peritonism, abdominal distension, absent bowel sounds, or bowel or omentum in the vagina or aspirate.** Suspect uterine perforation or bowel injury, which needs laparotomy.
- **Jaundice, dark urine, bronze skin, oliguria.** Suspect clostridial sepsis, haemolysis or acute kidney injury.
- **Bleeding from drip sites or gums, non-clotting blood.** Suspect DIC.
- **Closed os, empty uterus, positive pregnancy test and abdominal pain.** Think ectopic pregnancy.
- **Uterus larger than dates with heavy bleeding, hyperemesis or grape-like vesicles.** Think molar pregnancy.
- **Bradycardia with hypotension and tissue in the os.** Cervical shock: remove the tissue now.

Thinking like a clinician

- **Start with the os.**
 - **Closed os + uterus equals dates + viable pregnancy:** threatened.
 - **Open os:** the pregnancy cannot be saved (inevitable or incomplete).
 - **Closed os + small uterus:** either complete abortion, missed abortion or ectopic. **Scan before you conclude.**
- **"Complete abortion" is a diagnosis of evidence, not of hope.** Accept it when:
 - you saw a complete sac or products, **or**
 - a previous scan showed an IUP and the uterus is now empty, **or**
 - hCG falls appropriately. An **empty uterus without a previously confirmed IUP is a pregnancy of unknown location (PUL)** until proven otherwise.
- **Uterine size, not LMP, decides the method.** WHO bases incomplete-abortion treatment on **uterine size:** under 14 weeks vs 14 weeks or more. A woman at 16 weeks by dates whose uterus is 10-week size after passing most tissue can have MVA.
- **Every febrile woman with a recent pregnancy loss has septic abortion until proven otherwise.** Do not blame malaria alone. Do a malaria test, but still examine the pelvis.
- **Assume interference is possible in every case, without accusing anyone.** Signs such as tenaculum marks, cervical tears, foreign bodies, burns, or a woman who is very ill with an apparently "simple" miscarriage should change your plan: broader antibiotics, tetanus cover, a lower threshold for laparotomy.
- **Do not forget the second patient: her future.** Before she goes home, ask about her fertility intentions and whether she has a contraceptive method.
- **Common student mistakes:**
 - Doing an MVA on a shocked patient before resuscitation.
 - Missing an ectopic because the pregnancy test was "positive, so it must be an abortion".
 - Giving misoprostol for a threatened abortion with a live fetus.
 - Forgetting anti-D.
 - Discharging without FP.

- Not examining the removed tissue.

Clinical workup

This is the order in which a senior on the Mulago or Mengo gynae emergency ward works up a woman with **bleeding in early pregnancy**. Steps 0–3 happen within the first 15–30 minutes; she should not wait for a scan or a lab result before resuscitation or before POC are removed from the os.

Step 0: First 5 minutes

1. **Look at her from the door.** Is she pale, sweating, confused, gasping, lying in a pool of blood? If yes, **shout for help** (midwife, intern, anaesthetist) and go straight to resuscitation.
2. **Vitals now:** pulse, BP, respiratory rate, temperature, SpO₂, capillary refill, level of consciousness (AVPU).
3. **Calculate the shock index** (pulse ÷ systolic BP): **0.9 or more = abnormal, escalate; 1.4 or more = critical** (resuscitate and get blood). Remember young women keep their BP until late.
4. **Two large-bore cannulae (16–18 G);** draw blood for **Hb, grouping & crossmatch, bedside clotting test, mRDT, HIV and syphilis** at the same time.
5. **Start fluids** if shocked: **1 L warmed Ringer's lactate or normal saline quickly**, then reassess. Lie her flat, legs raised, oxygen if SpO₂ is low or she is shocked.
6. **Quick speculum at the bedside** if bleeding is heavy or she is bradycardic: **remove POC from the os with sponge forceps**. This is both diagnostic and treatment (cervical shock).
7. **Fever 38 °C or more, rigors or foul discharge?** Take cultures if possible but **give the first IV antibiotic doses within the hour**.
8. **Heavy bleeding and evacuation not immediately possible?** Give a uterotonic: **misoprostol 400 µg oral/sublingual or ergometrine 0.2 mg IM** (not if hypertensive).

RED FLAG

Call a senior at once for: **shock index 1.4 or more**, systolic BP under 90, **bradycardia with hypotension**, peritonism or distension, fever with hypotension or confusion, bleeding from drip sites, or **closed os with abdominal pain and signs of shock** (ruptured ectopic until proven otherwise).

Step 1: Focused history

Ask	Why (what changes the plan)
LMP and gestational age; any earlier scan showing an intrauterine pregnancy (IUP)?	A previously confirmed IUP makes ectopic much less likely. No previous scan + closed os + empty uterus = PUL, not "complete abortion".
Amount of bleeding: pads per hour, clots, "flooding"	1 pad or more soaked in under an hour = heavy; prepare for MVA and blood.
Tissue passed? Did she bring it?	Seeing a complete sac supports complete abortion; if she brought tissue, inspect it .
Pain type: midline cramps vs one-sided, shoulder-tip pain, fainting	Cramps = uterus expelling; unilateral pain or collapse = ectopic.
Fever, rigors, offensive discharge, days since the event	Sepsis: antibiotics first, evacuation within hours. Late presentation (3 days or more) raises sepsis risk.
Any interference (tablets, herbs, insertion, "a procedure")? Ask privately and gently.	Unsafe abortion: broaden cover, tetanus prophylaxis, look for perforation, foreign body or drug toxicity.
Previous CS or uterine surgery	Caution with repeated misoprostol at larger uterine sizes; perforation risk at MVA.

Ask	Why (what changes the plan)
Blood group if known, HIV status, sickle cell, bleeding disorder	Anti-D; ART linkage; transfusion planning.
Medical illness and allergies (asthma, hypertension, penicillin allergy)	Ergometrine contraindicated in hypertension; antibiotic choice.
Tetanus immunisation	Td ± tetanus immunoglobulin after an unsafe procedure.
Was the pregnancy wanted? Current contraception? Plans for future pregnancy?	Opens PAC counselling and guides the FP method you offer.

Step 2: Focused examination

- General:** pallor, jaundice (haemolysis, clostridial sepsis), dehydration, temperature, pulse, BP, RR, capillary refill, urine output. Document the **shock index**.
- Chest and heart** briefly: tachypnoea may mean sepsis, acidosis or ARDS.
- Abdomen:**
 - Is the uterus palpable above the symphysis (roughly 12 weeks or more)? Fundal height in weeks.
 - Tenderness, guarding, rebound, distension, absent bowel sounds** = peritonitis (perforation, bowel injury, ruptured ectopic, pelvic abscess).
- Speculum (the key step):**
 - Os open or closed?** Width of the os.
 - POC in the os or vagina?** Remove them now with sponge forceps and keep them to inspect.
 - Amount and **smell** of the blood; pus.
 - Cervical tears, tenaculum marks, burns, foreign bodies** (stick, catheter, herbs) = instrumentation.
 - Other sources: cervical polyp, ectropion, growth.
- Bimanual:**
 - Uterine size in weeks** (this, not LMP, decides MVA vs misoprostol), consistency, tenderness.
 - Cervical motion tenderness**, adnexal mass or tenderness, fullness in the pouch of Douglas.

Document the key negatives: "os closed/open, no POC in os, no offensive discharge, no cervical injury, uterus X weeks, non-tender, adnexae free, no peritonism".

Step 3: Bedside tests

Test	What it tells you
Urine pregnancy test (hCG)	Confirms pregnancy. May stay positive for 2–4 weeks after complete abortion, so a positive test does not mean retained tissue.
HemoCue / bedside Hb	Baseline; Hb under 7 g/dL or ongoing shock = transfuse.
Bedside clotting test	2 mL of blood in a plain glass tube held in the hand: no clot by 7 minutes , or a clot that breaks down, suggests DIC (sepsis, prolonged retention of a dead fetus).
mRDT for malaria	Fever plus pregnancy loss: treat malaria if positive, but still examine the pelvis for sepsis.
RBS	Septic or confused women; hypoglycaemia after quinine.
Urine dipstick	Nitrites/leucocytes (UTI); blood; ketones (dehydration).
Point-of-care ultrasound (if you are trained)	Free fluid in the abdomen or pouch of Douglas (ruptured ectopic or perforation), retained products, fetal heartbeat.
Inspect the tissue she passed or you removed	Float it in saline: villi look fluffy and fronded; a complete sac supports complete abortion; vesicles suggest mole.

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Hb / full blood count	Everyone (HC III+ for Hb; FBC at HC IV and hospitals)	Low Hb; WBC raised in sepsis; platelets under 100 suggest DIC	Transfusion; supports sepsis; triggers a clotting screen
Blood group, Rhesus D, crossmatch	Everyone (HC IV and above can crossmatch)	Rh-negative	Anti-D within 72 h after surgical evacuation or loss at 12 weeks or more; blood ready if heavy bleeding
Pelvic ultrasound (TVUS if available)	Unsure of the type, closed os, suspected missed abortion, ectopic, mole or perforation. Available at most HC IVs (abdominal) and all referral hospitals	Viable IUP (threatened); CRL 7 mm or more with no heartbeat or MSD 25 mm or more with no embryo (missed); heterogeneous retained products; empty uterus ± adnexal mass/free fluid; snowstorm (mole)	Confirms the type; an empty uterus without a prior IUP → serial hCG and ectopic pathway; below the cut-offs → rescan in 7–14 days
Serum β-hCG (quantitative)	PUL, suspected ectopic or mole. Mainly regional/national referral and private labs	Suboptimal rise or plateau (ectopic); very high (mole); falling by at least 50% in 48 h (failing pregnancy)	Decides discharge vs ectopic management
HIV and syphilis tests	Everyone (all levels)	Positive	ART and PMTCT linkage; benzathine penicillin; syphilis explains second-trimester loss
Blood culture, high vaginal swab	Septic abortion, before antibiotics if it causes no delay. Referral hospitals only in most of Uganda	Gram-negatives, anaerobes, <i>Clostridium</i>	Narrows antibiotics if she is not improving
Urea, creatinine, electrolytes	Sepsis, shock, oliguria, on gentamicin. HC IV (some) and hospitals	Rising creatinine = AKI	Adjust or stop gentamicin; fluid balance; dialysis referral
Clotting screen (PT, APTT, fibrinogen)	Septic shock, bleeding from sites, abnormal bedside test. Regional/national referral	Prolonged PT/APTT, fibrinogen under 2 g/L	Fresh frozen plasma, platelets; senior and anaesthetist involvement
Bilirubin, LFTs	Jaundice, suspected clostridial sepsis or haemolysis	Raised unconjugated bilirubin	Supports haemolysis; consider early hysterectomy if the uterus is necrotic
Erect chest X-ray / abdominal X-ray	Peritonism after an unsafe procedure	Gas under the diaphragm , foreign body	Laparotomy
Histology of tissue	Where available (referral hospitals), especially if mole suspected or no villi seen	Chorionic villi (confirms IUP); hydropic villi (mole)	Molar pregnancy → hCG follow-up; no villi → think ectopic

Step 5: Putting it together

Model one-line summary:

"A 26-year-old P2+1 at 11 weeks by dates with 1 day of heavy bleeding and cramps, **haemodynamically stable** (pulse 100, BP 110/70, shock index 0.9, afebrile), Hb 9.5 g/dL, with an **open os, POC in the os** and a **9-week-size non-tender uterus**. **Working diagnosis: incomplete abortion, not septic, no evidence of interference.**"

Problem list:

1. Incomplete abortion, uterine size under 14 weeks.
2. Mild-moderate anaemia (Hb 9.5 g/dL).
3. Rhesus status unknown.
4. Needs contraception (PAC).

Plan: POC removed from the os; MVA under paracervical block after ibuprofen (or misoprostol 600 µg orally if she prefers and can return); group and Rhesus; HIV/syphilis; iron; FP method before discharge.

Worked example 1: using the speculum to classify.

Speculum and bimanual findings	Classification	Next step
Os closed , small amount of blood from the os, uterus = dates, non-tender	Threatened (confirm a viable IUP on scan)	Reassure, paracetamol, return advice. No misoprostol, no MVA.
Os open , membranes bulging or no tissue passed yet, uterus = dates	Inevitable	Evacuate (MVA or misoprostol by uterine size).
Os open , tissue in the os, uterus smaller than dates	Incomplete	Remove POC; MVA or misoprostol 600 µg oral.
Os closed , bleeding settled, uterus small and firm, she saw a sac pass or a previous scan showed an IUP	Complete	No evacuation; PAC package.
Os closed , bleeding settled, uterus empty on scan, no prior IUP	PUL (not "complete")	Serial hCG 48 h apart; ectopic safety-net.
Os open, offensive pus , tender bulky uterus, fever	Septic	Antibiotics within 1 h, evacuate within hours.
Os open, tenaculum marks and a cervical tear , peritonism	Septic, induced, with possible perforation	Antibiotics, resuscitation, erect CXR, laparotomy.

Worked example 2: interpreting a scan report. A 30-year-old at 10 weeks by dates has spotting. Os closed. Scan: intrauterine sac, **MSD 18 mm, no embryo**.

- This is **below** the 25 mm cut-off, so you **cannot yet diagnose missed miscarriage** (dates may be wrong).
- Plan: **rescan in 7–14 days**. If she then has MSD 25 mm or more with no embryo, or an embryo with **CRL 7 mm or more and no heartbeat**, diagnose missed miscarriage and offer mifepristone + misoprostol, misoprostol alone, or MVA.
- Never give misoprostol "to be safe" on a single borderline scan: you might end a viable pregnancy.

Differential diagnosis

Condition	Distinguishing features	Key investigation
Ectopic pregnancy	Unilateral pain (often before the bleeding), shoulder-tip pain, fainting, cervical excitation, adnexal tenderness or mass. The os is closed and the bleeding is scanty and dark.	TVUS: empty uterus ± adnexal mass or free fluid; serial hCG
Molar pregnancy (gestational trophoblastic disease)	Uterus larger than dates, doughy; severe vomiting; early pre-eclampsia; passage of vesicles; theca-lutein cysts	Very high hCG; "snowstorm" appearance on ultrasound; histology of evacuated tissue
Cervical lesions (ectropion, polyp, cervicitis, cervical cancer)	Bleeding from a visible lesion; os closed; pregnancy normal	Speculum; scan confirms a viable IUP
Implantation bleeding	Light spotting around the expected period	Scan and dating
Septic abortion vs malaria / UTI / appendicitis	Uterine tenderness, offensive lochia and an open os point to the uterus	Speculum, pelvic exam, blood slide or mRDT, urinalysis, scan
Anovulatory or dysfunctional bleeding	Negative pregnancy test	Urine or serum hCG
Uterine perforation with bowel injury (after an unsafe procedure)	Peritonitis, distension, free gas, faeculent aspirate	Erect chest X-ray / abdominal ultrasound; laparotomy

Investigations

The full test list, with when to order each one and how it changes management, is in the **Step 4 table** of *Clinical workup* above. Three extra points:

- **Urine hCG may stay positive for 2–4 weeks** after a complete abortion. Do not use a positive test alone to diagnose retained tissue; persisting positivity beyond 3–4 weeks with symptoms suggests retained tissue, ectopic or gestational trophoblastic disease.
- In **septic abortion**, add **lactate** where available (a raised lactate supports septic shock) and repeat Hb and creatinine after 24 h.
- **Histology** is worth sending whenever the tissue looks abnormal (vesicles), no villi were seen, or there have been recurrent losses.

CLINICAL PEARL

Bedside clotting time. Put 2 mL of venous blood in a plain glass tube and hold it in your hand. If it has **not clotted by 7 minutes** (or clots and then breaks down easily), suspect coagulopathy. This works at any HC IV without a laboratory.

Management

Emergency algorithm (any woman with bleeding in early pregnancy)

1. **Call for help, ABC.** Lay her flat, raise the legs, give **oxygen** if in shock.
2. **Two large-bore IV cannulae** (16–18 G). Take blood at the same time for **Hb, group & crossmatch, malaria test, HIV and syphilis**, plus cultures and renal function if septic.

3. **Fluids:** warmed Ringer's lactate or normal saline, **1 L quickly**, then reassess pulse, BP and urine output. Transfuse if Hb is under 7 g/dL, or if she is still shocked after 1–2 L of crystalloid.
4. **Speculum: remove POC from the os** with sponge forceps. This often stops the bleeding and reverses cervical shock.
5. **Uterotonic** if bleeding is heavy and evacuation is not immediately possible: **ergometrine 0.2 mg IM** (avoid in hypertension) **or misoprostol 400 µg** orally or sublingually.
6. **Decide the type** (os, size, scan) and look for **sepsis or injury**.
7. **Evacuate the uterus** by the method that fits the uterine size and her condition (below).
8. **Antibiotics** if septic (IV) or after instrumentation.
9. **Anti-D, tetanus prophylaxis, analgesia.**
10. **PAC package before discharge:** counselling, **contraception**, HIV/STI services, and follow-up.

Threatened abortion

1. Scan to confirm an **intrauterine pregnancy with a heartbeat**, and check for a subchorionic haematoma.
2. Reassure: most (about 80–90% with a live embryo) continue normally.
3. **Paracetamol** for pain. Avoid NSAIDs in ongoing pregnancy.
4. **No evidence supports strict bed rest** or routine hormones. Advise her to avoid heavy work and intercourse until the bleeding stops, mainly for comfort.
5. **Do not give misoprostol or do MVA.** The fetus is alive.
6. Treat any cause you find: malaria, UTI, syphilis.
7. Tell her to **return** if the bleeding becomes heavy, pain increases, tissue passes or she develops a fever. Continue ANC.

Inevitable and incomplete abortion: uterine size under 14 weeks

Offer a choice where she is stable and not septic. All three options are effective.

Option	Details	Success	Best for
MVA (vacuum aspiration)	Paracervical block + NSAID; done in the procedure room	~98–100%	Heavy bleeding, sepsis (after antibiotics), wants it done now, unreliable follow-up
Misoprostol	600 µg orally, single dose, or 400 µg sublingually, single dose	~90–96% within 1–2 weeks	Stable woman, no sepsis, can return; HC III with no MVA provider
Expectant	Wait for spontaneous expulsion	Lower and slower	Stable, minimal bleeding, strongly prefers to wait

- Review **7–14 days after misoprostol**. Base the decision on her **symptoms and examination**, not on the scan alone. A thin rim of endometrial tissue or clot does not need MVA if she is well.
- **Do MVA if** bleeding is heavy, she is febrile, or retained products persist and she has symptoms.

EXAM FAVOURITE

"Why not D&C?" WHO recommends that **sharp curettage be replaced by vacuum aspiration**. D&C causes more blood loss, more pain, more perforation and more intrauterine adhesions (Asherman syndrome), and it usually needs general anaesthesia and a theatre.

Incomplete abortion: uterine size 14 weeks or more

1. Resuscitate; start **IV antibiotics** if any sign of infection.
2. **Misoprostol 400 µg sublingual, vaginal or buccal every 3 hours** until the fetus and placenta are expelled. WHO sets no fixed maximum, but use clinical judgement and **lower doses or caution with a previous**

uterine scar.

- Alternative when misoprostol is unavailable or the cervix is already open: **oxytocin 20–40 IU in 1 L normal saline or Ringer's lactate**, titrated to contractions (receptors increase as pregnancy advances).
- After expulsion:** examine the placenta. If it is incomplete or bleeding continues, remove the retained tissue with sponge forceps or MVA (large cannula). **Dilatation and evacuation (D&E)** should be done only by trained providers.
- Oxytocin 10 IU IM** after expulsion helps the uterus contract.

UGANDA CONTEXT

Ugandan randomised trials at district hospitals showed that **misoprostol 400 µg sublingually every 3 hours (up to 5 doses)** for second-trimester incomplete abortion achieved complete evacuation within 24 hours in over 90% of women, **with midwives as safe and effective as doctors**. MoH policy lets midwives manage first-trimester PAC. Second-trimester care has traditionally been reserved for physicians, but task sharing is expanding.

Missed abortion (early pregnancy loss with closed os)

Regimen	Details
Preferred (WHO 2022), under 14 weeks	Mifepristone 200 mg orally , then 1–2 days later misoprostol 800 µg vaginally, sublingually or buccally
Misoprostol alone (mifepristone not available)	Misoprostol 800 µg vaginally, sublingually or buccally; repeat doses may be given if needed (commonly 3-hourly, up to 3 doses)
Surgical	MVA under paracervical block (up to 12–14 weeks' uterine size, depending on the provider and cannula)
14–28 weeks (fetal death)	See the <i>Intrauterine fetal death</i> topic: mifepristone 200 mg, then misoprostol 400 µg every 4–6 hours

Complete abortion

- Confirm it (see *Thinking like a clinician*). If there was no previous scan showing an IUP, **scan, and consider serial hCG** to exclude ectopic.
- No evacuation needed. Give anti-D if Rh-negative (see below), analgesia, and the **PAC package**.

Septic abortion (a medical emergency)

- Resuscitate** (as in the algorithm).
 - Aim for MAP of 65 mmHg or more and urine output over 0.5 mL/kg/h.
 - Catheterise and keep a **fluid balance chart**.
 - Take **blood cultures and a high vaginal swab** if you can, but **do not delay antibiotics**.
- IV antibiotics within 1 hour:**

Drug	Dose	Route	Frequency	Notes
Ampicillin	2 g	IV	6-hourly	Covers streptococci, <i>Clostridium</i> , enterococci
Gentamicin	5 mg/kg	IV	Once daily (24-hourly)	Gram-negatives. Check renal function and hydration.
Metronidazole	500 mg	IV	8-hourly	Anaerobes

Drug	Dose	Route	Frequency	Notes
<i>Common alternative:</i> ceftriaxone + metronidazole	Ceftriaxone 1–2 g	IV	Once daily	Widely used in Ugandan hospitals. Still add metronidazole.

- **Continue IV until she has been fever-free for 48 hours**, then switch to oral **amoxicillin 500 mg 8-hourly** (or amoxicillin–clavulanate) + **metronidazole 400 mg 8-hourly** to complete **7–14 days**.
 - Base the switch and the total duration on her response.
3. **Evacuate the uterus promptly**, within a few hours of starting antibiotics, **by MVA**.
- Do not wait days for the fever to settle: **the retained tissue is the source**.
 - In large or second-trimester uteri, use misoprostol plus evacuation.
4. **Tetanus prophylaxis** after any unsafe procedure:
- **Tetanus toxoid (TT/Td) 0.5 mL IM**.
 - Plus **tetanus immunoglobulin 250 IU IM** if she is unimmunised or her status is unknown and the wound is dirty.
5. **Look for injury and act:**
- **Laparotomy** if there is peritonitis, suspected perforation or bowel injury, a pelvic abscess, or no improvement 24–48 hours after evacuation and antibiotics.
 - **Hysterectomy** may be lifesaving if the uterus is necrotic (e.g. clostridial infection) or there is uncontrollable haemorrhage.
6. **Supportive care:** analgesia, correct anaemia, watch for **DIC**, acute kidney injury and ARDS; involve anaesthesia / HDU.
7. **Refer** from HC IV to a regional or national referral hospital (e.g. Mulago, Kawempe) if she needs ICU, dialysis, or surgery beyond local capacity. **Start antibiotics and fluids before transfer**.

DRUG DOSING

Gentamicin 5 mg/kg once daily (e.g. 300 mg for a 60 kg woman). Reduce or stop it if urine output falls or creatinine rises, because it is nephrotoxic and ototoxic. In sepsis with acute kidney injury, ceftriaxone + metronidazole is a practical alternative.

Pain relief for PAC

Drug	Dose	Route	When
Ibuprofen	400–800 mg	Oral	30–60 min before MVA or with misoprostol. NSAIDs do not reduce misoprostol's effect.
Paracetamol	1 g	Oral	6–8-hourly, max 4 g/day
Paracervical block	Lidocaine 1% up to 20 mL (maximum 4.5 mg/kg without adrenaline)	Cervical injection	Before MVA (see <i>MVA kit</i> topic)
Diclofenac	75 mg	IM	Moderate to severe pain
Pethidine	50–100 mg	IM	Severe pain; watch for respiratory depression

WHO recommends **against routine general anaesthesia** for vacuum aspiration. A paracervical block plus an NSAID, with calm verbal support ("vocal local"), is enough for most women.

Rhesus (anti-D) prophylaxis

- **Rh-negative women having surgical evacuation**, and any Rh-negative woman losing a pregnancy **at 12 weeks or more**: give **anti-D immunoglobulin 250 IU IM** before 20 weeks (500 IU from 20 weeks), **within 72 hours**.
- WHO 2022 says **anti-D is not required for medical abortion under 12 weeks**.

UGANDA CONTEXT

Anti-D is expensive and often unavailable in public units. Check the Rhesus group anyway, document it, and if anti-D cannot be given, **tell her** so the next pregnancy is monitored for isoimmunisation.

PAC elements: the complete package

The PAC consortium model has **5 elements**:

1. **Community and service-provider partnerships.** Prevent unwanted pregnancy and unsafe abortion; mobilise women to seek care early.
2. **Counselling.** Address her emotional and physical health needs; explain what happened, the warning signs, and when she will be fertile again.
3. **Treatment** of incomplete and unsafe abortion and its complications: MVA, misoprostol, antibiotics, surgery.
4. **Contraceptive and family-planning services.** **Ovulation can return within 8–10 days**, so start contraception **before she leaves**.
5. **Reproductive and other health services:** HIV testing and linkage, syphilis and STI treatment, cervical cancer screening (VIA), nutrition and iron, screening for gender-based violence, and referral for infertility or recurrent-loss work-up.

Post-abortion family planning (WHO MEC)

Method	After 1st-trimester abortion	After 2nd-trimester abortion	After septic abortion
COC, POP, injectable (DMPA), implant	Start immediately (MEC 1)	Immediately (MEC 1)	Immediately (MEC 1)
Copper IUD / LNG-IUS	Immediately after MVA (MEC 1)	MEC 2 (higher expulsion)	Do not insert (MEC 4). Wait until the infection has resolved.
Condoms	Immediately; dual protection	Immediately	Immediately
Tubal ligation	Can be done at the time, if she chose it freely and was counselled before the emergency	Same	Delay until the infection has resolved
Fertility-awareness methods	Delay until regular cycles return	Delay	Delay

CLINICAL PEARL

If she gets misoprostol, an **implant or injectable can be given the same day**. An IUD can go in once the abortion is confirmed complete at follow-up. A Ugandan study found that offering an IUD after medical management of incomplete abortion is acceptable to many women. Do not make her come back twice.

What to do where

Level	What you can do
HC III	Diagnose; resuscitate (IV fluids); remove POC from the os; misoprostol for incomplete abortion under 14 weeks; MVA by a trained midwife or clinical officer; first-dose antibiotics; FP; refer shocked, septic or second-trimester cases
HC IV	All of the above plus MVA , second-trimester misoprostol, blood transfusion , laparotomy (perforation, ectopic), IV antibiotics
Regional / national referral	ICU / HDU, dialysis, hysterectomy for necrotic uterus, bowel surgery, D&E, specialist ultrasound, management of complex complications

Before any referral: IV line and fluids running, first antibiotic dose given, uterotonic if bleeding, POC removed from the os, a referral note with **vitals and drugs given**, and a health worker escorting her.

Follow-up

- **7–14 days** after misoprostol, or 1–2 weeks after MVA.
- Check: bleeding, pain, fever; contraception satisfaction; results (HIV, syphilis); histology.
- **Pregnancy test** at 3–4 weeks only if symptoms persist. Persistent positive hCG suggests retained tissue, ectopic, or gestational trophoblastic disease.

Complications

Complication	Clues	Action
Haemorrhage / hypovolaemic shock	Tachycardia, hypotension, ongoing bleeding	Resuscitate, remove POC, uterotonic, evacuate, transfuse
Retained products	Persistent bleeding, open os, fever	Repeat MVA or misoprostol
Sepsis / septic shock / DIC	Fever, tachycardia, hypotension, bleeding from sites	Antibiotics, evacuation, supportive care, blood products
Uterine perforation	Instrument passes further than expected; peritonitis	Observe if small and she is stable; laparotomy if bowel injury or bleeding
Bowel or bladder injury	Faeculent aspirate, peritonitis, haematuria	Laparotomy, surgical team
Tetanus	Trismus, spasms, days after an unsafe procedure	Prevent with TT ± TIG; ICU care
Acute kidney injury	Oliguria, rising creatinine (sepsis, haemolysis, gentamicin)	Fluids, stop nephrotoxins, dialysis referral
Long-term	Asherman syndrome (adhesions), secondary infertility (tubal damage), chronic pelvic pain, ectopic in future, isoimmunisation, psychological distress	Counselling, FP, fertility referral

Ward-round & exam pearls

Model summary line:

"Mrs N, a 24-year-old P1+1 at 10 weeks by dates, presented with 2 days of per vaginal bleeding and lower abdominal cramps, with passage of clots. She is **haemodynamically stable** (pulse 96, BP 112/70, afebrile), Hb 9.8 g/dL. On speculum the **os is open with products in the os**, and the uterus is 8-week size, non-tender. **Impression: incomplete abortion, not septic**. Plan: POC removed from the os; MVA under paracervical block with ibuprofen; group and Rhesus; HIV test; **post-abortion family planning counselling, she has chosen an implant**; review before discharge."

Viva questions and answers:

1. *What decides MVA vs misoprostol?* Uterine size (under 14 weeks either works), her stability (heavy bleeding or sepsis favours MVA), her preference, and whether she can come back.
2. *What is cervical shock and how do you treat it?* A vagal response to POC stretching the cervix. Bradycardia and hypotension out of proportion to blood loss. Remove the POC with sponge forceps; lie her flat; give fluids.
3. *Name the 5 elements of PAC.* Community partnerships, counselling, treatment, contraception/FP, other reproductive health services.
4. *Why start IV antibiotics before evacuating a septic abortion, but evacuate soon?* Antibiotics reduce bacteraemia during instrumentation. Evacuation removes the infected necrotic tissue that antibiotics cannot sterilise.
5. *When can she start an IUD after abortion?* Immediately after first-trimester MVA (MEC 1), unless there is septic abortion (MEC 4).
6. *What is the WHO misoprostol dose for incomplete abortion under 14 weeks?* 600 µg orally or 400 µg sublingually, single dose.
7. *A woman bleeds in early pregnancy, the os is closed and the uterus is empty on scan. Diagnosis?* Pregnancy of unknown location. It could be a complete miscarriage, a very early IUP or an ectopic. Do serial hCG and do not discharge without a safety net.

EXAM FAVOURITE

OSCE tip: In a PAC counselling station, cover:

- what happened, and that it was not her fault if spontaneous;
- danger signs: fever, heavy bleeding, foul discharge, severe pain;
- fertility returns within 2 weeks;
- her contraceptive choice (offer all methods with pros and cons);
- HIV testing;
- follow-up date.

Keep your tone warm and non-judgemental; examiners mark it.

Mnemonics & memory aids

"TIC-MS": the types in the order the os and the uterus decide them

- Threatened: **T**ight (closed) os, **T**ruer-to-dates uterus, heartbeat present
- Inevitable / Incomplete: os **I**s open
- Complete: **C**losed, **C**ontracted small uterus, **C**alm (bleeding stopped)
- Missed: **M**ute (no symptoms), **M**acerated or dead embryo, closed os
- Septic: **S**mell, **S**hivers, **S**ore uterus

The septic abortion antibiotic trio: "A-G-M" (like "a German") Ampicillin 2 g 6-hourly + Gentamicin 5 mg/kg daily + Metronidazole 500 mg 8-hourly, until 48 hours fever-free.

Misoprostol "6-4 under 14; 4 every 3 over 14"

- Under 14 weeks: **600 µg** oral or **400 µg** sublingual, once.
- 14 weeks or more: **400 µg** every **3** hours.

PAC = "C-C-T-F-R" Community, Counselling, Treatment, Family planning, Reproductive health services.

High-yield summary

Numbers & doses to know

Item	Value
Abortion definition (Uganda / WHO)	Before 28 weeks / before 22 weeks or under 500 g
Shock index	0.9 or more abnormal; 1.4 or more critical
Transfusion trigger	Hb under 7 g/dL , or still shocked after 1–2 L crystalloid
Missed miscarriage on scan	CRL 7 mm or more with no heartbeat, or MSD 25 mm or more with no embryo
Incomplete abortion, uterus under 14 weeks	MVA, or misoprostol 600 µg oral once or 400 µg SL once
Incomplete abortion, uterus 14 weeks or more	Misoprostol 400 µg SL/vaginal/buccal every 3 h until expulsion
Missed abortion under 14 weeks	Mifepristone 200 mg orally, then misoprostol 800 µg 1–2 days later (or misoprostol 800 µg alone)
Uterotonic while awaiting evacuation	Misoprostol 400 µg oral/SL or ergometrine 0.2 mg IM
Septic abortion antibiotics	Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily + metronidazole 500 mg IV 8-hourly
IV antibiotics until	Fever-free 48 h , then oral to complete 7–14 days
Paracervical block	Lidocaine 1% up to 20 mL ; max 4.5 mg/kg (plain)
Tetanus after unsafe abortion	Td 0.5 mL IM ± TIG 250 IU IM
Anti-D (Rh-negative)	Within 72 h ; UK 250 IU before 20 weeks; the Ugandan vial usually stocked is 1,500 IU
Bedside clotting test	No clot by 7 minutes = coagulopathy
Follow-up after misoprostol	7–14 days
Return of ovulation	As early as 8–10 days

Classic exam traps

- **"Positive hCG + bleeding = miscarriage."** Wrong: closed os + empty uterus with no previous IUP is a **PUL**, possibly ectopic. Do serial hCG.
- **"Bradycardia and hypotension, so give more fluids and wait for theatre."** Wrong: this is **cervical shock**. Remove POC from the os with sponge forceps now.
- **"Septic abortion: give antibiotics for 48 h to cool her off, then evacuate."** Wrong: start antibiotics, then **evacuate within hours**; the retained tissue is the source.
- **"Use the LMP to decide MVA vs misoprostol."** Wrong: **uterine size** on bimanual decides (under 14 vs 14 weeks or more).

- **"Fever after an abortion is malaria."** Wrong until you have **examined the pelvis**: septic abortion first.
- **"Oxytocin is the best drug for a first-trimester incomplete abortion."** Wrong: few oxytocin receptors before about 20 weeks; use **misoprostol**.
- **"Insert an IUD at the time of MVA for septic abortion."** Wrong: **MEC 4**. Offer an implant, injectable or pills instead.
- **"A small retained rim on the day-10 scan needs MVA."** Wrong if she is well: manage by **symptoms and examination**, not the scan alone.

Questions seniors ask

- **What is the first thing you do for a woman bleeding heavily at 10 weeks?** Assess ABC and vitals, calculate the shock index, two large-bore cannulae with bloods, fluids, and a speculum to remove POC from the os.
- **How do you tell inevitable from incomplete abortion?** Both have an open os; in incomplete, **some tissue has already passed** and the uterus is usually smaller than dates.
- **What makes you suspect an induced (unsafe) abortion?** Tenaculum marks, cervical tears, burns, foreign bodies, a very sick woman after an apparently "simple" miscarriage, or peritonism.
- **When would you do a laparotomy in septic abortion?** Peritonitis, suspected perforation or bowel injury, pelvic abscess, gas under the diaphragm, or no improvement 24–48 h after antibiotics and evacuation.
- **What are the scan criteria for missed miscarriage?** CRL 7 mm or more with no heartbeat, or MSD 25 mm or more with no embryo; below that, rescan in 7–14 days.
- **What must she leave with?** A contraceptive method (if she wants one), HIV/syphilis results or linkage, anti-D if Rh-negative and indicated, danger-sign advice and a follow-up date.
- **When is an empty uterus reassuring?** Only if a previous scan showed an IUP, or she passed a sac you saw, or hCG falls appropriately.

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define abortion as used in Ugandan clinical practice.	Loss or termination of pregnancy before fetal viability , taken as before 28 weeks in Uganda (WHO uses <22 weeks or <500 g for international comparison).
2	Threatened vs inevitable abortion: the single distinguishing sign?	The cervical os: closed in threatened (pregnancy may continue), open in inevitable (pregnancy cannot be saved).
3	Features of incomplete abortion	Heavy bleeding, cramping, open os , products may be seen in the os, uterus smaller than dates , some tissue already passed.
4	Features of missed abortion	Dead embryo/fetus retained; loss of pregnancy symptoms, brownish spotting or none, closed os, uterus smaller than dates ; scan: CRL ≥ 7 mm without heartbeat or MSD ≥ 25 mm without embryo.
5	What is cervical shock and its treatment?	Vagal reflex from products of conception distending the cervical canal → bradycardia + hypotension out of proportion to blood loss . Treat by removing the POC from the os with sponge forceps , lie flat, IV fluids.
6	WHO 2022 misoprostol regimen for incomplete abortion, uterine size <14 weeks	600 µg orally single dose or 400 µg sublingually single dose. (MVA is the alternative.)
7	WHO misoprostol regimen for incomplete abortion ≥ 14 weeks' uterine size	400 µg sublingual, vaginal or buccal every 3 hours until expulsion (caution/lower doses with a uterine scar).
8	WHO preferred regimen for missed abortion <14 weeks	Mifepristone 200 mg orally , then 1–2 days later misoprostol 800 µg vaginal/sublingual/buccal. Misoprostol alone 800 µg is the alternative.
9	Septic abortion: first-line IV antibiotic trio (WHO/Uganda)	Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV 24-hourly + metronidazole 500 mg IV 8-hourly until fever-free 48 h, then oral to complete 7–14 days.
10	Why must a septic abortion be evacuated promptly rather than after days of antibiotics?	Retained necrotic infected tissue is the source ; antibiotics cannot sterilise it. Evacuation = source control . Start antibiotics first, then evacuate within hours.
11	Organism causing haemolysis, bronze skin, gas in tissues and renal failure after unsafe abortion	<i>Clostridium perfringens</i> (gas gangrene). May need hysterectomy.
12	Two extra prophylactic measures after an unsafe (instrumented) abortion	Tetanus toxoid 0.5 mL IM ($\pm$ tetanus immunoglobulin 250 IU if unimmunised) and anti-D if Rh-negative (where available).
13	Name the 5 elements of post-abortion care (PAC)	1. Community & service-provider partnerships 2) Counselling 3) Treatment of incomplete/unsafe abortion & complications 4) Contraceptive/FP services 5) Reproductive & other health services (HIV, STI, cervical screening, GBV).
14	How soon can ovulation return after an abortion?	As early as 8–10 days – so contraception should be started before discharge .
15	Can an IUD be inserted immediately after MVA for incomplete abortion?	Yes after first-trimester abortion (MEC 1); MEC 2 after second trimester; NOT after septic abortion (MEC 4) – delay until infection resolved.

#	Question	Answer
16	Why is D&C (sharp curettage) no longer recommended?	WHO: replace with vacuum aspiration. D&C has more blood loss, pain, perforation, Asherman syndrome , usually needs GA/theatre. MVA is safer and can be done by midwives.
17	Maximum dose of lidocaine for paracervical block (plain)	4.5 mg/kg (about 200 mg = 20 mL of 1% for most adults). Aspirate before injecting.
18	Bleeding in early pregnancy, closed os, EMPTY uterus on scan, no previous IUP seen – diagnosis?	Pregnancy of unknown location – could be complete miscarriage, very early IUP or ectopic . Serial hCG ± repeat scan; safety-net advice.
19	Why is oxytocin a poor drug for first-trimester uterine evacuation?	The uterus has few oxytocin receptors before ~20 weeks ; prostaglandins (misoprostol) act regardless of gestation.
20	Is PAC legal in Uganda?	Yes . Uganda's abortion law is restrictive and contested, but post-abortion care is legal, lifesaving and a patient right – never withhold or delay it, keep confidentiality.
21	Uterotonic options for heavy bleeding in incomplete abortion while awaiting evacuation	Ergometrine 0.2 mg IM (avoid in hypertension) or misoprostol 400 µg oral/SL ; remove POC from os; resuscitate.
22	Shock index: how is it calculated and what are the obstetric cut-offs?	Pulse ÷ systolic BP. 0.9 or more = abnormal, escalate; 1.4 or more = critical, resuscitate and get blood. BP falls late in young women.
23	Ultrasound criteria for diagnosing missed miscarriage	CRL 7 mm or more with no heartbeat , or mean sac diameter 25 mm or more with no embryo . Below these, rescan in 7–14 days before acting.
24	Speculum signs that an abortion was instrumented (unsafe)	Tenaculum marks , cervical tears, burns, foreign bodies (stick, catheter, herbs), offensive discharge. Change the plan: broad antibiotics, tetanus prophylaxis, look for perforation.
25	Bedside clotting test: how to do it and what is abnormal?	2 mL venous blood in a plain glass tube , held in the hand. No clot by 7 minutes (or a clot that breaks down) = coagulopathy (think DIC in septic abortion).
26	Closed os, bleeding settled, empty uterus on scan, no previous scan. Is this a complete abortion?	No: it is a pregnancy of unknown location . Accept 'complete' only if a prior scan showed an IUP, a sac was seen passing, or hCG falls appropriately. Do serial hCG 48 h apart and give ectopic safety-net advice.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 23-year-old P0+1 at 9 weeks presents to a HC IV with light bleeding and mild cramps. Pulse 84, BP 118/74, afebrile. Speculum: closed os, minimal blood. Ultrasound: intrauterine gestational sac with an embryo, CRL 22 mm, heartbeat present. What is the MOST appropriate management?

- A Reassurance, paracetamol, return precautions and continue ANC
- B Manual vacuum aspiration
- C Misoprostol 400 µg sublingually
- D Admit for strict bed rest and progesterone injections
- E Ergometrine 0.2 mg IM

2. A 28-year-old P2+0 at 11 weeks by LMP has had heavy bleeding with clots for 6 hours. Pulse 48, BP 80/50, pale and sweaty. Estimated visible blood loss is moderate. On speculum, tissue is seen protruding through the os. What is the MOST appropriate immediate action?

- A Give atropine and wait for the pulse to recover
- B Give misoprostol 600 µg orally and observe
- C Transfer immediately to theatre for laparotomy
- D Remove the products from the os with sponge forceps while resuscitating
- E Perform an urgent ultrasound scan before anything else

3. A 19-year-old student presents 3 days after 'a procedure' at a private clinic for an unwanted pregnancy of 10 weeks. Temperature 39.2 °C, pulse 124, BP 88/56, offensive vaginal discharge, tender bulky uterus, open os. Abdomen is soft apart from suprapubic tenderness. After starting IV fluids, what is the BEST next sequence?

- A Oral amoxicillin and metronidazole, review in 1 week
- B IV ampicillin, gentamicin and metronidazole now, then MVA within hours once antibiotics are running
- C IV antibiotics for 48–72 hours until afebrile, then MVA
- D Immediate MVA before any antibiotics to remove the source
- E Laparotomy and hysterectomy

4. A 32-year-old P3+0 had MVA for septic incomplete abortion yesterday and is on ampicillin, gentamicin and metronidazole. Today she has worsening abdominal distension, absent bowel sounds, generalised rebound tenderness and the MVA aspirate is noted to have smelt faeculent. Erect CXR shows gas under the diaphragm. What is the MOST appropriate management?

- A Continue antibiotics and repeat MVA
- B Add IV ceftriaxone and observe 48 hours
- C Nasogastric tube and conservative management of ileus
- D Refer for CT scan before deciding
- E Resuscitate and perform laparotomy

- 5.** A 25-year-old at 8 weeks' uterine size has an incomplete abortion. She is stable (pulse 88, BP 120/76, afebrile, Hb 11.2 g/dL), bleeding is moderate, and she lives 2 km away with a phone. The HC III has misoprostol but the only MVA-trained midwife is on leave. What is the MOST appropriate treatment?
- A Refer to the regional hospital for D&C
 - B Misoprostol 400 µg sublingually every 3 hours until expulsion
 - C Misoprostol 600 µg orally as a single dose with review in 7–14 days
 - D Oxytocin 20 IU infusion
 - E Expectant management with no follow-up
- 6.** A 30-year-old P2+1 at 17 weeks by dates presents with heavy bleeding; fetus was expelled at home 2 hours ago. Uterus is 16-week size, os open, placenta not delivered. Pulse 100, BP 110/70, afebrile. Which is the MOST appropriate uterotonic regimen to complete the evacuation?
- A Misoprostol 600 µg orally single dose
 - B Misoprostol 25 µg vaginally 6-hourly
 - C Misoprostol 400 µg sublingually every 3 hours until the placenta is expelled, then check for retained tissue
 - D Mifepristone 200 mg and discharge
 - E Ergometrine 0.5 mg IV every 15 minutes
- 7.** A 22-year-old presents with vaginal bleeding at 7 weeks by LMP. Urine hCG positive. The bleeding is scanty and dark, and she reports left iliac fossa pain that started 2 days before the bleeding. Speculum: closed os. Pulse 104. What is the MOST important diagnosis to exclude first?
- A Ectopic pregnancy
 - B Complete abortion
 - C Threatened abortion
 - D Cervical ectropion
 - E Molar pregnancy
- 8.** A woman had MVA for first-trimester incomplete abortion. She is afebrile, not septic, and wishes to avoid pregnancy for at least 3 years. Which contraceptive option is MOST appropriate to offer before she leaves the procedure room?
- A Advise her to wait 6 weeks before starting any method
 - B Emergency contraception only
 - C Fertility-awareness method starting now
 - D Tubal ligation immediately without prior counselling
 - E Copper IUD inserted immediately after the MVA
- 9.** A 26-year-old had septic incomplete abortion evacuated today. She wants an IUD. What is the MOST appropriate advice?
- A Insert a copper IUD now to reduce future infection
 - B Insert an LNG-IUS now
 - C IUDs are contraindicated for life after septic abortion
 - D Defer IUD until infection has resolved; offer an implant or injectable now
 - E Only condoms are allowed for 6 months

10. A 34-year-old G4P3 at 12 weeks is found on scan to have an embryo with CRL 30 mm and no cardiac activity. She has only brown spotting and a closed os. She is stable and requests medical management. Mifepristone is available. Which regimen is BEST? Advanced

- A Misoprostol 600 µg orally once
- B Mifepristone 200 mg orally, then misoprostol 800 µg vaginally/sublingually/buccally 1–2 days later
- C Misoprostol 25 µg vaginally 4-hourly
- D Oxytocin infusion
- E Ergometrine 0.2 mg IM daily for 3 days

11. A 21-year-old presents with fever 38.8 °C 4 days after passing tissue at home at 9 weeks. The clinician finds a positive malaria RDT and plans to discharge her on artemether-lumefantrine. What is the MOST important omission?

- A Checking her HIV status
- B Speculum and bimanual examination for septic abortion
- C Giving iron supplements
- D Repeating the malaria test by microscopy
- E Checking her blood group

12. A 29-year-old P1+0 is Rhesus D-negative and has MVA for incomplete abortion at 11 weeks. Anti-D is available. What should you do? Advanced

- A No anti-D is ever needed before 12 weeks
- B Give anti-D only in the next pregnancy
- C Give anti-D 1500 IU IV only if Kleihauer is positive
- D Give anti-D 250 IU IM within 72 hours
- E Give anti-D only if the partner is Rh-positive and she consents to paternity testing

13. At an HC III, a 20-year-old with septic incomplete abortion at 10 weeks has pulse 130, BP 80/40, confusion and oliguria. The facility has no theatre, blood or IV gentamicin, but has IV fluids, ceftriaxone and metronidazole. What is the BEST action?

- A Start IV fluids, give IV ceftriaxone and metronidazole, remove POC from os, and refer urgently with an escort
- B Perform MVA at the HC III immediately and keep her there
- C Refer immediately without starting any treatment to save time
- D Give oral amoxicillin and observe overnight
- E Give misoprostol 600 µg and send home

14. Which finding on history/examination MOST strongly suggests that an incomplete abortion was induced with instrumentation?

- A Uterus smaller than dates
- B Passage of clots
- C Tenaculum marks and a laceration on the anterior lip of the cervix
- D Lower abdominal cramping
- E Positive urine hCG

15. A 24-year-old at 10 weeks' uterine size with incomplete abortion receives misoprostol 600 µg orally. At review on day 10 she feels well, bleeding is light, she is afebrile, and ultrasound shows a 12 mm heterogeneous endometrial echo. What is the MOST appropriate management? Advanced

- A Immediate MVA because of retained products
- B Repeat misoprostol 600 µg and review in 1 week
- C Order serum hCG daily for a week
- D Start IV antibiotics
- E Reassure – treatment successful; no further intervention; confirm contraception

Answers & explanations

1. A · Reassurance, paracetamol, return precautions and continue ANC

This is a **threatened abortion** (closed os, live IUP). Correct: **reassure, analgesia (paracetamol), safety-net advice**, continue ANC; most continue normally. **B/C** would end a live pregnancy – absolutely wrong. **D**: strict bed rest and routine progestogens have no proven benefit in threatened miscarriage and hospital bed rest carries VTE and cost burdens. **E**: ergometrine contracts the uterus and could provoke expulsion; it has no role with a live fetus.

2. D · Remove the products from the os with sponge forceps while resuscitating

Bradycardia with hypotension out of proportion to bleeding and tissue at the os = **cervical shock** (vagal). **Remove the POC with sponge forceps** – usually reverses it within minutes – together with IV fluids and lying flat. **A**: atropine may be used if bradycardia persists but does not treat the cause. **C**: no evidence of intraperitoneal bleeding. **B**: misoprostol takes hours; the os is already open with tissue sitting there. **E**: scanning delays a lifesaving 30-second manoeuvre.

3. B · IV ampicillin, gentamicin and metronidazole now, then MVA within hours once antibiotics are running

Septic abortion with septic shock: **IV broad-spectrum antibiotics immediately, then prompt evacuation (source control) within hours**. **A**: oral outpatient therapy is dangerous in septic shock. **C**: waiting days leaves infected necrotic tissue in place – a classic audit failure in Uganda. **D**: evacuating before antibiotics increases bacteraemia; antibiotics take minutes to start. **E**: laparotomy/hysterectomy is for peritonitis, perforation, necrotic uterus or failure to respond – not first-line here.

4. E · Resuscitate and perform laparotomy

Peritonitis + free gas + faeculent aspirate after an unsafe procedure = **uterine perforation with bowel injury** → **resuscitate and laparotomy** (with surgical team; may need bowel repair/stoma ± hysterectomy). **A/B/C** treat symptoms while bowel contents soil the peritoneum. **D**: CT is unnecessary and unavailable in most units; the diagnosis is clinical + X-ray, and delay is lethal.

5. C · Misoprostol 600 µg orally as a single dose with review in 7–14 days

Stable, non-septic incomplete abortion <14 weeks: **misoprostol 600 µg PO single dose** (or 400 µg SL) is WHO-recommended with ~90–96% success; review at 7–14 days. **A**: D&C is obsolete and referral is unnecessary. **B**: the 3-hourly repeat regimen is for ≥14 weeks' uterine size. **D**: oxytocin works poorly in the first trimester (few receptors). **E**: expectant management is an option but must include follow-up and safety-netting.

6. C · Misoprostol 400 µg sublingually every 3 hours until the placenta is expelled, then check for retained tissue

Uterine size ≥ 14 weeks: **misoprostol 400 µg SL/PV/buccal every 3 hours** until expulsion, then inspect placenta and remove remaining tissue. **A**: single-dose 600 µg is for <14 weeks. **B**: 25 µg is a third-trimester labour induction dose – far too low here. **D**: mifepristone alone is slow and discharge with a retained placenta and heavy bleeding is unsafe. **E**: wrong dose/route; ergometrine standard is 0.2 mg IM/IV and it can trap the placenta.

7. A · Ectopic pregnancy

Unilateral pain **preceding** scanty dark bleeding, closed os and tachycardia = **ectopic pregnancy until proven otherwise** (the fatal miss). **B/C** are possible but less dangerous and the history fits ectopic better. **D**: ectropion causes painless contact bleeding. **E**: molar pregnancy usually gives uterus larger than dates and hyperemesis, not unilateral pain.

8. E · Copper IUD inserted immediately after the MVA

After uncomplicated first-trimester abortion, **immediate IUD insertion is MEC category 1** and ideal for long-acting protection. **A**: ovulation may return within 8–10 days – waiting 6 weeks exposes her to repeat pregnancy. **C**: fertility-awareness methods are unreliable until cycles are regular. **D**: sterilisation requires prior informed, unpressured counselling – not decided in an emergency. **B**: EC is not ongoing contraception.

9. D · Defer IUD until infection has resolved; offer an implant or injectable now

IUD after **septic abortion is MEC 4** – defer until infection resolves; meanwhile **hormonal methods (implant, DMPA, COC) can start immediately (MEC 1)**. **A/B** risk worsening pelvic infection. **C**: the restriction is temporary. **E**: all hormonal methods are safe now.

10. B · Mifepristone 200 mg orally, then misoprostol 800 µg vaginally/sublingually/buccally 1–2 days later

Missed abortion <14 weeks: WHO recommends **mifepristone 200 mg then misoprostol 800 µg 1–2 days later** (more effective than misoprostol alone). **A**: 600 µg single dose is the incomplete-abortion regimen; missed abortion needs higher dose as the os is closed and tissue fully attached. **C**: third-trimester induction dose. **D/E**: ineffective in early pregnancy.

11. B · Speculum and bimanual examination for septic abortion

Any fever after pregnancy loss is **septic abortion until proven otherwise** – examine the pelvis (offensive discharge, open os, tender uterus). Malaria can co-exist. **A/C/E** are good PAC components but not the dangerous omission. **D**: repeating the malaria test does not address the likely uterine source.

12. D · Give anti-D 250 IU IM within 72 hours

Surgical evacuation in an Rh-negative woman: **anti-D 250 IU IM within 72 h** (before 20 weeks). WHO says anti-D is not needed for *medical* abortion <12 weeks, but instrumentation increases fetomaternal haemorrhage. **A** overgeneralises. **C**: Kleihauer is used after 20 weeks to size the dose. **B**: prophylaxis must be given now. **E**: impractical and not standard – treat all Rh-negative women.

13. A · Start IV fluids, give IV ceftriaxone and metronidazole, remove POC from os, and refer urgently with an escort

Septic shock with organ dysfunction needs a higher-level facility, but **stabilise and start antibiotics before transfer**: IV fluids, first-dose antibiotics, remove POC from the os, then urgent escorted referral with a note. **B**: she needs HDU care, blood and theatre backup. **C**: untreated transfer is a common preventable cause of death. **D/E** are dangerously inadequate.

14. C · Tenaculum marks and a laceration on the anterior lip of the cervix

Tenaculum marks/cervical lacerations (and foreign bodies, vaginal burns) indicate instrumentation – raising risk of sepsis, perforation and tetanus. **A/B/D/E** occur in spontaneous miscarriage too.

15. E · Reassure – treatment successful; no further intervention; confirm contraception

Success after misoprostol is judged **clinically**: she is well with light bleeding and a thin (<15 mm) echo is expected – blood/decidua, not necessarily retained products. **A/B** overtreat and expose her to unnecessary procedures. **D**: no signs of infection. **C**: unnecessary; hCG testing at 3–4 weeks only if symptoms persist.

Clinical cases

Work through each step before reading the model answer.

Case 1: Heavy bleeding at 10 weeks at a HC IV

A 27-year-old P2+0 is brought to a HC IV in Mukono at 8 pm with heavy per vaginal bleeding and clots for 5 hours at about 10 weeks by LMP. She has soaked 4 pads in the last hour and feels dizzy.

1. What are your immediate priorities?

Call for help; **ABC**; lie flat, legs raised, oxygen; **2 large-bore IV cannulae**; draw blood for **Hb, grouping & crossmatch, malaria test, HIV/syphilis**; **1 L warmed Ringer's/NS** rapidly; vitals every 15 min; catheter for urine output.

2. What focused history do you take while resuscitating?

LMP and early scan; amount of bleeding and **tissue passed**; pain character (crampy vs unilateral/shoulder-tip); fever/foul discharge; **any interference** (asked privately, non-judgementally); previous CS; contraception and pregnancy intention; Rh group; HIV status; tetanus immunisation.

3. Pulse 118, BP 92/60, afebrile, Hb (HemoCue) 8.1 g/dL. Speculum: open os with tissue in the canal; uterus 8-week size, non-tender, no adnexal masses. What is the diagnosis and the immediate step?

Incomplete abortion with haemorrhagic shock (compensating), no sepsis. **Remove POC from the os with sponge forceps** immediately; give **misoprostol 400 µg SL** or **ergometrine 0.2 mg IM** (if normotensive) to contract the uterus; continue fluids; crossmatch 1–2 units.

4. What definitive treatment and supporting care?

MVA once resuscitation is under way: **ibuprofen 400–800 mg PO** (if able to swallow) and **paracervical block (lidocaine 1%, ≤20 mL)**; appropriate cannula; examine the tissue (villi confirm IUP). Transfuse if Hb <7 or still unstable. **Anti-D** if Rh-negative. Oral iron.

5. Before discharge, what does the complete PAC package include?

Counselling (what happened, danger signs, fertility returns within 2 weeks); **contraception chosen and started** (e.g. implant or immediate IUD – MEC 1); **HIV test and linkage**, syphilis screen; iron; **follow-up in 1–2 weeks**; community link (VHT).

Take-away: Resuscitate and clear the os first; MVA is definitive; PAC is not complete until she leaves with contraception and follow-up.

Case 2: Fever after an unsafe abortion

An 18-year-old S.4 student is brought by her aunt to a regional referral hospital with fever, lower abdominal pain and foul-smelling discharge 4 days after she 'removed a pregnancy' of about 12 weeks with the help of a local herbalist.

1. What history is most important and how will you ask it?

Alone with her, with privacy and assurance of confidentiality: what was done (instruments, sticks, herbs, drugs – e.g. chloroquine, pessaries), when, bleeding, tissue passed, pain (worsening/generalised?), vomiting, diarrhoea, urine output, jaundice. Tetanus vaccination. LMP, contraception, sexual history (consent – consider defilement; she may be a minor; safeguarding).

2. Temp 39.5 °C, pulse 132, BP 84/50, RR 28, cold extremities, jaundiced. Abdomen: distended, generalised guarding, absent bowel sounds. Speculum: offensive discharge, open os, a laceration of the cervix. What is your assessment?

Septic abortion with septic shock, probable **uterine perforation with peritonitis** (possible bowel injury); jaundice suggests **haemolysis (clostridial?)** or hepatic dysfunction/sepsis. Differentials for the jaundice: severe malaria, drug toxicity.

3. What investigations?

FBC, **grouping/crossmatch**, blood culture, HVS, **urea/creatinine/electrolytes, bilirubin/LFTs, clotting (or bedside clotting time)**, malaria test, HIV, urinalysis, **erect CXR (free gas)**, abdominal/pelvic ultrasound (retained POC, free fluid, collections).

4. Outline management step by step.

1. Resuscitate: oxygen, 2 IV lines, crystalloid boluses, catheter, fluid chart. 2) **IV ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily (adjust for renal function) + metronidazole 500 mg 8-hourly** (or ceftriaxone + metronidazole) immediately. 3) **Tetanus toxoid 0.5 mL IM + TIG 250 IU**. 4) Blood products for DIC. 5) **Laparotomy** (peritonitis + suspected perforation) with surgical team: repair perforation/bowel, peritoneal lavage, evacuate uterus; **hysterectomy** if uterus necrotic/gangrenous. 6) HDU/ICU postoperatively.

5. What complications should you anticipate post-operatively and what follow-up?

Septic shock, DIC, AKI (sepsis, haemolysis, gentamicin), ARDS, wound sepsis, pelvic abscess, tetanus, death. Later: infertility, chronic pelvic pain. Follow-up: psychosocial support, **contraception** (implant/DMPA now; no IUD while septic), HIV/STI care, school re-entry support, safeguarding referral if a minor.

Take-away: Unsafe abortion kills through sepsis and hidden visceral injury – antibiotics, source control (evacuation or laparotomy) and tetanus cover, delivered without judgement.

Case 3: Painless bleeding: is it really a miscarriage?

A 31-year-old G3P2 at 7 weeks by LMP presents to Mengo casualty with 3 days of dark spotting. She says she passed 'something' yesterday. Pregnancy test positive. She has mild left lower abdominal discomfort.

1. What are the possibilities and what examination findings would help?

Complete, incomplete, threatened or missed miscarriage; **ectopic pregnancy**; molar pregnancy. Examine: vitals (tachycardia, postural drop), abdominal tenderness/guarding, speculum (os open/closed, tissue), bimanual (cervical excitation, adnexal mass, uterine size).

2. Vitals normal. Os closed, minimal old blood, uterus bulky, mild left adnexal tenderness, no mass felt. TVUS: empty uterus, endometrium 9 mm, no adnexal mass, no free fluid. No previous scan. Diagnosis?

Pregnancy of unknown location (PUL) – NOT 'complete abortion', because an IUP was never documented and she has adnexal tenderness. She did not bring the tissue.

3. What is your next investigation plan?

Serum hCG now and at 48 hours. Fall >50% suggests failing pregnancy (complete miscarriage); rise >63% suggests an IUP (rescan when above discriminatory level); a suboptimal rise/plateau or slow fall suggests ectopic. Group and Rh, Hb.

4. hCG 1,850 IU/L, and 48 h later 1,920 IU/L; she now has more left-sided pain. What does this mean and what will you do?

A plateauing hCG above the discriminatory zone with an empty uterus = likely ectopic pregnancy. Admit, repeat TVUS by an experienced sonographer, prepare for laparoscopy/laparotomy or methotrexate if criteria met (see Ectopic topic). Do not evacuate the uterus blindly.

5. What should she be told if she had been discharged earlier?

Safety-net advice: return immediately with worsening one-sided pain, shoulder-tip pain, dizziness/fainting or heavy bleeding; attend for repeat hCG. Document clearly.

Take-away: Never call it 'complete abortion' without evidence of an IUP or passed sac – an empty uterus is a PUL, and ectopic must be excluded.

References

1. Uganda Clinical Guidelines 2023: National Guidelines for Management of Common Health Conditions. *Ministry of Health Uganda*, 2023. <https://www.differentiatedservicedelivery.org/wp-content/uploads/UCG-2023-Publication-Final-PDF-Version-1.pdf>
2. Abortion care guideline – Incomplete abortion management: Recommendations 35–38. *World Health Organization*, 2022. <https://srhr.org/abortioncare/chapter-3/post-abortion-3-5/incomplete-abortion-management-recommendations-35-38-3-5-2/>
3. Abortion care guideline – Recommendation 31: Medical management of missed abortion <14 weeks. *World Health Organization*, 2022. <https://srhr.org/abortioncare/chapter-3/abortion-3-4/clinical-services-recommendation-31-medical-management-of-missed-abortion-at-gestational-ages-14-weeks-3-4-3/>
4. Clinical practice handbook for quality abortion care. *World Health Organization*, 2023. https://cdn.who.int/media/docs/default-source/reproductive-health/clinical-practice-handbook-for-quality-abortion-care.pdf?sfvrsn=e82e253c_7
5. FIGO's updated recommendations for misoprostol used alone in gynecology and obstetrics. *Int J Gynecol Obstet* (Morris et al.), 2017. <https://obgyn.onlinelibrary.wiley.com/doi/full/10.1002/ijgo.12181>
6. A randomized trial of misoprostol compared with manual vacuum aspiration for incomplete abortion (Kampala). *Obstet Gynecol* (Weeks et al.), 2005. <https://pubmed.ncbi.nlm.nih.gov/16135584/>
7. Treatment of incomplete second-trimester abortion with misoprostol by midwives and physicians: randomised equivalence trial in Uganda. *BMJ Global Health / PMC*, 2022. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9605879/>
8. Management of abortion complications at a rural hospital in Uganda: a criterion-based audit. *BMC Women's Health*, 2015. <https://link.springer.com/article/10.1186/s12905-015-0233-y>
9. Antibiotic susceptibility profiles and determinants of septic abortion among women admitted for abortion care in Ugandan satellite teaching hospitals. *Health Science Reports*, 2026. <https://onlinelibrary.wiley.com/doi/10.1002/hsr.272062>
10. Barriers and facilitators in the provision of post-abortion care at district level in central Uganda. *BMC Health Services Research / PMC*, 2014. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3903434/>

11. Uptake of intrauterine contraception after medical management of first trimester incomplete abortion in central Uganda. *PLOS One*, 2025. <https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0310936>
12. Septic Abortion. *MSD Manual Professional Edition*. <https://www.msdmanuals.com/professional/gynecology-and-obstetrics/early-pregnancy-disorders/septic-abortion>
13. Misoprostol for Treatment of Incomplete Abortion: An Introductory Guidebook. *Gynuity Health Projects*. https://gynuity.org/assets/resources/clinguide_pacguide_en.pdf

A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.